

A collaborative approach to the evaluation of cognitive accessibility of a conversational agent for public healthcare

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Abstract

Conversational agents are increasingly being introduced in healthcare to enhance access to healthcare and address limited healthcare resources. The aim of this study was to explore experiences of a conversational agent for symptom checking and triage tool assessing the urgency and appropriate level of care implemented in public healthcare from a cognitive accessibility perspective.

With a Participatory Action Research approach, this study includes: (1) A Participatory Cognitive Barrier Walkthrough in workshops with people with cognitive impairment ($n = 7$), product owner representatives ($n = 4$), and researchers ($n = 5$), followed by a workshop for improvement suggestions; (2) a paired interview with healthcare system specialists ($n = 2$).

The data from the study were analyzed using reflexive thematic analysis, resulting in five themes: ‘The design of the eHealth service created expectations that were not met during interaction’; ‘The eHealth service uses a difficult language’; ‘The eHealth service triggers negative emotions and reactions’; ‘The layout and graphic design of the eHealth service is difficult and sometimes misleading’; and ‘The eHealth service lacks in trustworthiness’.

The findings show the importance of including people with lived experience of cognitive impairment, to expose and find solutions for cognitive accessibility issues.

RESEARCH HIGHLIGHTS

- This study presents a collaborative method to influence design when system ownership is external to research.
- Involving people with impairments in eHealth design and evaluation improves cognitive accessibility.
- Design and language in the evaluated system affected user’s expectations, trust, emotions, and experience of cognitive accessibility.

Keywords: interaction design; Ehealth; participatory action research; process and methods; participatory design; user evaluation

1 Introduction

Conversational agents are increasingly introduced into healthcare (Tudor Car et al., 2020) as a mean to increase access to healthcare. Digital and online symptom checkers that provide healthcare advice and guidance on course of action to the user are also introduced in healthcare, as a means to release the increasing burden on healthcare services (Chambers et al., 2019; Wallace et al. 2022). By 2050, the global population over 60 will nearly double from 12% in 2015 to 22% (World Health Organization, 2022a). This demographic shift will significantly increase the demand for healthcare services due to age-related conditions. The World Health Organization (2022b) currently estimates that 16%

of the world population have impairments that affect their life to such extent that they experience a disability (World Health Organization, 2022b). Conversational agents and digital and online symptom checkers have the potential to contribute to solve some of the problems with universal health coverage. For sustainable development, it is reasonable to develop eHealth that is accessible and fits the needs of the most frequent healthcare users, that is elderly people and people with impairments (Gulley and Altman, 2008). The ongoing demographic changes with an increasing population of elderly people is expected to increase the prevalence of people with cognitive impairment (Karen et al., 2022). Also, The Convention on the Rights of Persons with Disabilities (CRPD) demand States Parties that to enable persons with disabilities

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to live independently and participate fully in all aspects of life, appropriate measures must be taken to ensure access to information and communications technologies and systems (United Nations, 2006) e.g., eHealth.

In line with the demands, the aim of this study was to explore aspects of cognitive accessibility of an online symptom checker and triage tool for assessing the urgency and appropriate level of care presented as a conversational agent implemented in public healthcare. The exploration was conducted through a collaborative approach involving end-users and other relevant stakeholders.

2 Background

2.1 Accessibility

Accessibility is defined as:

‘extent to which products, systems, services, environments and facilities can be used by people from a population with the widest range of characteristics and capabilities to achieve a specified goal in a specified context of use’ (International Organization for Standardization, 2020).

Accessibility specifically addresses the user needs and experiences of people with impairments. To increase accessibility, the European Union (EU) formed a directive that mandates member states public sector bodies’ websites and apps to comply with a minimum level of accessibility (Directive 2016/2102) described in the standard EN 301549 (European Telecommunications Standards Institute: ETSI, 2018). The standard builds on the international Web Content Accessibility Guidelines (WCAG) 2.1 (World Wide Web Consortium, 2018). WCAG is a technical standard that has been criticized for not providing enough guidance on cognitive accessibility (Seeman, 2006; Small et al., 2005; Miesenberger et al., 2019). The International Organization for Standardization (ISO) developed a general guideline on cognitive accessibility to be used in design and development of systems, products and built environments, the ISO 21801-1:2020 (International Organization for Standardization, 2020). However, standards and guidelines fail to see accessibility as situated in the context where the specific webpage or app is used for the intended purpose (Johansson, 2019, p. 42). As a complement to standards and guidelines, it is imperative that end users are involved in both development and evaluation of cognitive accessibility of the digital tools they are supposed to use.

2.2 Conversational agents

Conversational agents are described by Tudor Car et al. (2020) as ‘computer programs designed to simulate human text or verbal conversations’. Conversational agents are called by different names in research, for example virtual assistants (Guzman, 2019), chatbots (Apuzzo and Buresi, 2022; Mugoye et al., 2019; Parmar et al., 2022), and smart-bots (Mugoye et al., 2019). Sometimes the term used for the conversational agent reveals the purpose of the chatbot, e.g., healthbot (Parmar et al., 2022).

Conversational agents in healthcare have been introduced in a wide range of areas, e.g., maternal healthcare (Mugoye et al., 2019), COVID-19 (Amiri and Karahanna, 2022; White et al., 2022), mental health (Cameron et al., 2019), post-surgery of hip arthroscopy (Dwyer et al., 2023), and promoting health behavior changes (Aggarwal et al., 2023). The potential of universal access to healthcare through conversational agents is exemplified by the WHO that has provided several publicly available conversational agents (World Health Organization, 2023). One of them is the virtual healthcare worker Florence.

Florence can have conversations about tobacco products, healthy living, COVID-19 vaccines, and mental health in six languages (World Health Organization, 2021).

There is currently lack of consensus regarding accessibility guidelines for chatbots according to Stanley et al. (2022). Stanley et al. (2022) conducted a review and found 17 sources for chatbot accessibility and present a list of 157 unique recommendations. However, 108 of the recommendations did not reference user studies or existing guidelines and are described as based on reasoning about the user experience (Stanley et al., 2022). The authors also pointed out that visual impairment was the most commonly addressed impairment. Stanley et al. (2022) propose that more empirically validated recommendations are needed. The authors stated that future research should address under-researched populations such as people with dyslexia, cognitive impairment, mental health conditions, mobility impairments, and impaired hearing. Further on, Stanley et al. (2022) proposed several future research directions e.g., investigating accessibility testing methods and resources, and accessibility considerations for users with multiple impairments. In a literature review published in 2024, Huq et al. (2024) identified eight conversational agents with machine learning or deep learning method involving health issues or chronic conditions that address the disabilities and helplessness of elderly and people with cognitive disorders. However, this review did not address accessibility explicitly. In a scoping review of conversational agents for adults with brain-related neurological conditions, Hocking et al. (2023) stated that barriers and facilitators for use and safety were poorly reported in the included studies. The review did not describe any studies regarding cognitive accessibility or accessibility in general.

The advances in natural language processing and AI have presented various kinds of conversational agents. Several publicly available generative pretrained transformer (GPT) and large language models (LLM) exist since the launch of ChatGPT in November 2022. These technologies can generate conversations, both written and in synthesized voice, that closely mimics humans. The presence of conversational agents in society is increasing. The technology in these conversational agents is evolving rapidly. This emerging technology has several implications for healthcare, including chatbots that provide medical suggestions and counselling based on symptoms and medical background (Javaid et al., 2023). However, the ChatGPT technology was not ready to be implemented in healthcare according to a study published in April 2023 (Au et al., 2023).

2.3 Digital and online symptom checkers

Digital and online symptom checkers can target specific or general populations as well as specific or general conditions (Chambers et al., 2019). Digital and online symptom checkers can be developed using diverse technologies and designs (e.g., questionnaire or chatbot). A chatbot-based symptom checker (CSC) is a conversational agent that can offer a human-like conversation to inform the user with recommendations and/or a medical diagnosis. The system in the conversational agents vary from rule-based fixed questions and answers to comprehending artificial intelligence (AI) that learn from and adapt questions and answers depending on the conversations (Tudor Car et al., 2020). The conversation can rely on different methods as text, image, and voice (Tudor Car et al., 2020). Previous research shows that evaluations of conversational agents in healthcare focus on accuracy, effectiveness, and acceptability (Tudor Car et al., 2020). There is also previous research regarding accuracy in digital

and online symptom checkers (Chambers et al., 2019; Wallace et al., 2022).

Savolainen and Kujala (2024) conducted usability testing of two online symptom checkers used in Finland. The testing focused on cognitive accessibility and challenges for certain groups at risk of digital exclusion. The symptom checkers consisted of text-based questionnaires that users need to answer to be assisted to appropriate care, with AI assessment of the urgency of care. Three groups of participants were recruited: people with mild intellectual disability, older adults, and non-native Finnish speakers. The results revealed that numerous challenges were encountered by all participants. The authors report that some issues were pointed out at almost every test session, and stress the importance of simplifying services, shortening lengthy lists of symptoms, and using understandable terminology (Savolainen and Kujala, 2024).

To the best of our knowledge, there is no research focusing on how design can affect the experience of cognitive accessibility in CSC.

3 Method

3.1 Study rational and overall methodological approach

This study had a descriptive study design with a qualitative approach consisting of two consecutive data collections, described in detail in the Procedure section.

This study is part of the CoDeAc research program (CoDeAc, 2020). CoDeAc conduct research aiming to design eHealth services that everyone can use. CoDeAc work from the idea that those who face the greatest difficulties should have the most influence over how the technology is designed. CoDeAc has participatory action research (PAR) (Macdonald, 2012) and participatory design (PD) (Frauenberger et al., 2015) as theoretical and methodological frameworks. Value sensitive design (VSD) (Friedman et al., 2013) is guiding on human values and ethics to be accounted for in the process of shaping the technology. PAR aims to be a democratic way of conducting research together with the people affected by the problem and acting for change in the society (Macdonald, 2012). PD is a collaborative design approach involving end users in the design process (Frauenberger et al., 2015). PAR and PD share many core values, such as democratizing processes and empowering the participants (Frauenberger et al., 2015), but PD focus more on the mutual learning between designers and people in a context (Bødker et al., 2022).

This study was initiated by the research program CoDeAc and set out to conduct research on an already implemented eHealth service in public healthcare, the '1177 direkt'. The ownership of the targeted system was outside the research program. The second author invited participants from the product owner to the study. The activities in the first part of the data collection of the study had a collaborative approach, involving stakeholders with ownership of the system together with participants from the research program CoDeAc to explore cognitive accessibility. The product owner of 1177 direkt, the Inera company, provided access to the eHealth service. This study was conducted in the spirit of PAR and PD but is a mix of the four co-design directions described by Mattelmäki and Sleeswijk Visser (2011). We describe the mix as a collaborative approach, making a specific user group's voice heard for part of the design process in co-design activities together with stakeholders from the system owner.

To evaluate the 1177 direkt, we used a Participatory Cognitive Barrier Walkthrough (PCBW) method developed in the CoDeAc research program. This method draws from accessibility

evaluation methods presented by Brajnik (2008). According to Brajnik (2008), it is important to consider context when evaluating accessibility. That is to be specific on what type of impairment people have and what they are trying to achieve, their goals, when using a specific resource on the internet. In PCBW, focus is on users with cognitive impairments. The method moves away from individual user testing scenarios and from isolated expert evaluations on accessibility conformance. Instead, both people with cognitive impairments and accessibility experts are collaborating in a series of workshops applying user scenarios in a walkthrough setting with the goal to identify barriers and to jointly discuss the severeness of a specific barrier or a combination of barriers.

The second part of the data collection explored user experiences from real-life setting related to the results from the PCBW. For this part, a semi-structured paired interview was conducted with healthcare system specialists.

3.2 Study context

Sweden provides public healthcare for a low fee through tax-funded healthcare carried out by 21 self-governing regions and municipalities. The regions and municipalities cooperate through the Swedish Association of Local Authorities and Regions (SALAR¹, 2023a).

The Swedish Health and Medical Services Act ('Hälsö och sjukvårdslag, SFS, 2017:30) that regulates public healthcare is based on an ethical platform with three principles ('1996/97:SoU14', 1997). These principles declare that people have equal value and the same rights, regardless of personal characteristics and societal functions, that people with the greatest needs should be prioritized, and that healthcare should be organized to promote cost-effectiveness (Hälsö och sjukvårdslag, SFS, 2017:30).

All regions provide general health and healthcare information and personal information and eHealth services through the Swedish national web-portal 1177.se. The regions also provide healthcare advice by the telephone number 1177. To access personal information and eHealth services the user needs to login with an approved electronic identification. In the year 2023, Sweden had a population of 10.6 million people (Statistics Sweden, 2024). There are 9.7 million resident accounts registered for personal eHealth services in 1177.se with around 12 million monthly logins for eHealth services and 15 million monthly visits to the web portal for general health and healthcare information (Swedish Association of Local Authorities and Regions, 2023b).

The software system in the national web-portal and eHealth services and the telephone services are carried out by the Inera company (2022), owned by SALAR. The provided eHealth services are often acquired through public procurement, which means that a private company may be the product owner.

In 2023, a CSC and triage tool was launched by Inera in an eHealth service called '1177 direkt'. Triage is a process in healthcare that assesses patients to determine the urgency and appropriate level of care of the health condition and recommends the patient to the appropriate level of healthcare. The CSC and triage tool's medical content are based on the 'Rådgivningsstödet', a clinical decision support for nurses (Taube, 2023). Nine healthcare regions in Sweden implemented 1177 direkt. The word 'direkt' in the name means immediately. An important note is that the 1177 direkt may be occupied with different systems due

¹ In Sweden known as SKR, the Swedish abbreviation of SALAR.

TABLE 1. Participants in the study, their affiliation, and attendance in workshops or the paired interview.

Participant ID	Attendance at workshop (WS) or paired interview (PI)	Affiliation
P1	WS 1-4	Begripsam group
P2	WS 1-4	Begripsam group
P3	WS 2-3	Begripsam group
P4	WS 1-4	Begripsam group
P5	WS 1-3	Begripsam group
P6	WS 1-4	Begripsam group
P7	WS 1-4	Begripsam group
P8	WS 2-4	Inera
P9	WS 1-4	Inera
P10	WS 3	Inera
P11	WS 1-4	Platform24
P12	WS 1-4, PI	Researcher, KTH Royal Institute of Technology
P13	WS 1,2,4	Researcher, Center for Clinical Research Dalarna, Uppsala University
P14	WS 1,3,4	Researcher, KTH Royal Institute of Technology
P15	WS 4	Researcher, Center for Clinical Research Dalarna, Uppsala University
P16	WS 4	Researcher, KTH Royal Institute of Technology
P17	PI	Healthcare system specialist from public healthcare organization
P18	PI	Healthcare system specialist from public healthcare organization

to Inera procuring several suppliers. In this study, the supplier was Platform24, used by nine regions at the time of the study. The supplier Platform24 also delivered the CSC and a triage tool to at least one region outside of Inera's public procurement under another name. In this paper, we will refer to the systems delivered by Platform24 as 1177 direkt.

A person living in one of the nine regions that offer 1177 direkt can reach the eHealth service through a webpage or an app. The chatbot will triage the patient either to self-care advice, to contact a healthcare professional or to call the number 112 for emergency care. If a patient is referred to a healthcare professional, they may either wait for availability or be required to book an appointment for online consultation. The healthcare provider decides the process for patients; therefore, the process may vary for different healthcare providers. If the latter method is chosen, the healthcare provider must allocate available time slots in a calendar with a set access time limit, allowing patients to book appointments. Depending on the number of patients seeking care and being triaged to healthcare professional, the waiting time and availability of appointments may vary. Before the consultation, the healthcare professional receives a system generated summary text of the patient's answers from the automated chat and information about the triage outcome, i.e., priority of urgency regarding the health condition. The healthcare professional will also have access to the original questions and answers from the chat. The healthcare professional can text chat with the patient and initiate a video consultation if necessary.

3.3 Participants

A team of evaluators was formed for the workshops, consisting of seven members from the Begripsam group, five researchers, and four persons representing the product owner perspective from Inera and Platform24.

Participants were given specific roles before the workshops. Members from the CoDeAc acted as evaluators and co-designers. Participants from Inera and Platform24 acted as observers and were invited to co-design in the dedicated co-design workshop. The first author acted as workshop leader, a designated researcher

(third or last author) took notes, and all researchers participated actively in the evaluation and co-design activities.

The Begripsam group is a member organization that work for accessibility in the society, with a special focus on cognitive accessibility. Information about the study and an invitation to participate in the study was distributed to members of the Begripsam group. Two male and five female participants in ages between 27 and 67 years were enrolled in the study. All seven participants had previous experience of evaluating digital systems with the method used in this study. All participants from the member organizations had impairments relating to neuropsychiatric, energy/drive, executive, or memory. In addition, two had neurological and musculoskeletal impairments, and four had impairments relating to communication, language, or calculation.

The researchers, two male and three female researchers between 31 and 61 years, represented the fields of human computer interaction, medicine and health, and implementation science. Altogether, the researchers have expertise in accessibility, design, usability, healthcare practice, and the Swedish healthcare system infrastructure. The researchers are also the authors of this paper.

The two healthcare system specialists were recruited to a semi-structured paired interview as a purposeful sample to get the user perspective from healthcare system specialists. The participants had deep knowledge of 1177 direkt and practical experience on implementing the system at one of the public healthcare providers in a Swedish region. They were involved in the implementation process that included handling user evaluations from patients and healthcare professionals. Therefore, these participants had a broad knowledge of reported user experiences from both healthcare professionals and patients.

An overview of participants' attendance at workshops and paired interview is presented in Table 1.

Prior to the study all participants gave written consent to participate after been given written and oral information, along with an opportunity to ask questions and get additional information about the study. Ethical approval was granted by the Swedish Ethical Review Authority, reference number 2019-04419.

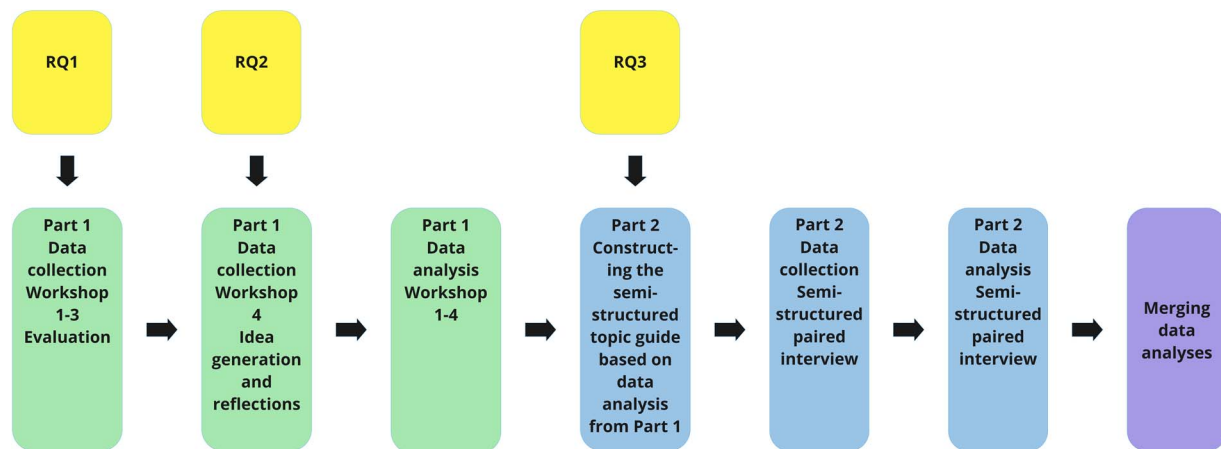


FIGURE 1. A process map of the conducted research activities in seven steps. RQ1 is connected to the first step, evaluation workshops. RQ2 is connected to the second step, idea generation workshop. RQ3 is connected to step four, constructing the semi-structured topic guide.

3.4 Procedure

The study consists of two parts: (1) four workshops and (2) one paired interview. Data Collection and data analysis were undertaken in a stepwise procedure shown in Figure 1 in the following order: (1 and 2) data collection at workshops, (3) analysis of data from the workshops, (4) constructing the semi-structured topic guide, (5) data collection for the paired interview, (6) analysis of data from the paired interview, and (7) merging the analyses from workshops and the paired interview (Figure 1). To validate that our findings relate to cognitive accessibility, the codes from workshops 1–3 were linked to the requirements in the cognitive standard ISO 21801-1:2020 (International Organization for Standardization, 2020) during analysis of data from workshops in step 2.

This study seeks to answer the following research questions:

RQ1: How do people with cognitive impairment experience an automated eHealth service related to cognitive accessibility?

RQ2: How can people with cognitive impairment and other stakeholders be engaged in co-design for idea generation to solve identified accessibility issues?

RQ3: How do healthcare system specialists responsible for implementation of an automated eHealth service in public healthcare experience the user perspective?

To explore these research questions, we focused on the eHealth service 1177 direkt. We used the PCBW method for RQ1, co-design for RQ2, and semi-structured paired interview for RQ3.

3.5 Preparations and scaffolding the workshops

A pre-meeting was held for participants from Inera and Platform 24 to inform about the online workshop method, as a part of the scaffolding all the workshops and to foster a creative and safe workshop environment. The instructions included talking slowly, using common words or explain terms, and ensuring that only the researchers provided information about how the 1177 direkt worked. The latter approach was constructed for the evaluation to mimic the experience for patients using the 1177 direkt for the first time, without previous knowledge of the 1177 direkt and relying on public information. The different roles were explained, and participants were encouraged to view all critique as valuable learning opportunities and to refrain from explaining design features in the system.

Notes taken during the workshops were posted on a Miro board and could consist of written text, drawings, and screenshots from the application. A link to the Miro board was shared with all participants.

During the first workshop, participants introduced themselves, and detailed information about the research project, the workshop agenda, and upcoming workshops was provided. To help participants recall previous activities, workshops 2–4 began with participants reintroducing themselves and reviewing the activities from the previous workshops. The ‘raise hand’ function in Zoom was used to indicate a desire to speak, and the workshop leader facilitated the discussion by passing the speaking turn to participants. This approach ensured a clear speaking order and prevented simultaneous talking. If someone used uncommon words or terminology, the workshop leader either asked the participant to explain the term or provided an explanation. When needed, the workshop leader reminded participants to speak more slowly. One-to-one support was offered to participants from the Begripsam group with technology or systems, for example the Miro-board, if needed. Participants were offered various ways to comment or reflect after workshops, using the Miro-board or direct contact with the researchers by phone or email. These approaches were also part of the scaffolding techniques.

3.6 Evaluation workshops

We conducted a total of four video-recorded online workshops in Zoom to explore the user experience of 1177 direkt. Focusing on cognitive accessibility, we conducted three evaluation workshops of the 1177 direkt with PCBW method in-group sessions. The scenario in the PCBW was participants with lived experience of cognitive impairments evaluating the conversational agent 1177 direkt with a test case during video-recorded workshop sessions. During the PCBW, the participants discussed their experiences, detected barriers and accessibility issues, and proposed improvements. The evaluation workshops lasted between 90 and 120 min with a 10-min break at half time. All three workshops were conducted in the same manner. Addressing the entire process for one test case required three workshops. Each subsequent workshop began from the point where the previous one concluded. In the first workshop, participants were introduced to 1177 direkt with the public information available on the Swedish national health information website 1177.se. The test case was a health condition described

for participants as a fictive person stating: 'I have a headache. It started today. It is not very painful. Sometimes I get tension headaches'. The workshop leader shared screen of the 1177 direkt and used the automated symptom check and triage process based on instructions and choices made by participants from Begripsam group. The automated symptom checker and triage process was considered as completed when the interaction with the conversational agent came to an end. At this point, the conversational agent guided the patient to one of the three recommendations: self-care, a healthcare professional, or emergency services.

3.7 Co-design workshop

A fourth workshop was dedicated to a co-design session on improvements, lasting for 120 min including a break. This workshop was prepared by the first, second, and last author who identified important accessibility issues from the previous workshops to be used as starting points for discussions on improvements.

After reintroductions and recap of the first three workshops along with reflections, the workshop leader presented a range of identified accessibility issues and informed participants that a pre-selected set would be used for the co-design session. All participants were encouraged to actively engage. Product owner representatives were specifically instructed to contribute, build on ideas, and indicate if an idea was unfeasible to allow for new ideas to be generated. The accessibility issues were discussed until one or more idea was generated to solve the issue. The pre-selected accessibility issues are presented in the first column of Appendix 2.

3.8 Semi-structured paired interview

A semi-structured paired interview was conducted by the first author with two participants in a video meeting in Zoom that was recorded. The participants received written information about the study and a written consent form before the interview meeting. The meeting started with participants introducing themselves, a presentation of the study, repeating information about the procedure including that the interview would be video recorded after the participants had orally consented to participate. The video recording was transcribed verbatim to text, after which the recording was deleted.

3.9 Data collection

In this study, we considered design aspects in the investigated eHealth service as 'the entire process across the full range of domains required for any given outcome' (Redström, 2017, p. X), consisting of both all the details in the 1177 direkt such as words, layout, buttons, information, and the process to complete the task in the 1177 direkt. During the video-recorded workshops, discussions were facilitated to be about how the design and language in the 1177 direkt was experienced and suggestions of improvements.

A semi-structured topic guide was constructed in Swedish, informed by the findings in the workshops, and was used for the paired interview (English translation, Appendix 1). The topic guide included questions about the participants' own experiences of the design; healthcare professionals' and patients' reported experiences of design; how the participants thought the design affected patients, professionals, and the quality of healthcare; and suggestions for improvement. During the video-recorded pair interview, the moderator posed probing and follow-up questions, restated and summarized information, and asked the participants to confirm the accuracy of the spoken data. In line with

Wilson et al. (2016), the semi-structured paired interview aimed to collect information about shared experiences, concerns, and needs from the respective participant's perspectives unpacking both similarities and differences. It followed a constructionist conception approach, in that the knowledge is co-constructed by all participants, generating situated meanings around the topics.

3.10 Data analysis

A reflexive thematic analysis (Braun and Clarke, 2019) was conducted in an iterative process following the six steps presented by Braun and Clarke (2006): (1) familiarizing yourself with your data, (2) generating initial codes, (3) searching for themes, (4) reviewing themes, (5) defining and naming themes, and (6) producing the report.

The data analysis was conducted in two phases. The first phase was an inductive analysis conducted independently for the two data sets from workshops and paired interview, steps 1–3. The analysis was predominantly semantic, where the latent approach was about interpreting the underlying meaning of feelings when feelings were expressed with laughter, frowning, sarcasm, or indirect expressions indicating a feeling. After the independent analysis, the two data analyses were merged, steps 4–6. In the first stage, data were extracted from the recordings of the workshops and paired interview and coded by the first author. Data extraction from the workshops consisted of transcriptions of meaning units on sticky notes together with attached related screenshots. The recording from the paired interview was transcribed verbatim, meaning units presented on sticky notes together with initial codes. Meaning units with verbatim transcriptions were presented on sticky notes together with initial code on a Miro board. Examples of the coding process are presented in Table 2.

The second phase of the analysis was theoretically driven to explore if the codes from the workshops related to cognitive accessibility using the ISO 21801-1:2020 guidelines on cognitive accessibility. The codes on the Miro board were marked with specific guideline numbers derived from the ISO 21801.

For a deeper understanding of the data the codes, the themes, and the codes' relation to the guidelines on cognitive accessibility were analyzed, discussed, and elaborated in an iterative process by first, second, fourth, and last authors. The video recordings and data were revisited several times, as were coding, and allocating codes into themes and reviewing themes.

All data collection was in Swedish. Translations of quotes and in figures were conducted with Google translate and Microsoft Copilot and edited when necessary to preserve the original meaning or essence.

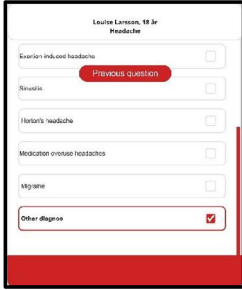
4 Result

The participants expressed many barriers to cognitive accessibility during the 25 steps the interaction lasted. The result is presented in five themes developed from the data:

- The design of the eHealth service created expectations that were not met during interaction.
- The eHealth service uses a difficult language.
- The eHealth service triggers negative emotions and reactions.
- The layout and graphic design of the eHealth service is difficult and sometimes misleading.
- The eHealth service lacks in trustworthiness.

All figures in the result section are visualizations created by first author, showing the main content on the specific page that we wish to highlight for the purpose in the result. There may

Table 2. Examples of coding process of the data from the workshops and the paired interview.

	Workshop	Paired interview
Theme	4. The layout and graphic design of the eHealth service is difficult and sometimes misleading	difficult and sometimes misleading
Examples of codes	The response options are illogical	The response options come in illogical order
Example of meaning units	'If you go back up and look at the instruction text, so I think what can make it difficult is that it says more options come in the next question. Then you can wonder are there more headache diagnoses that come in the next question too? If so, it's a bit strange that you have to say "other diagnosis" here if you haven't seen them all.'	'And very often the patient's reaction to that is: why did this question come up now? I also think that is important.'
A translated visualization of the screenshot on the Miro board		N/A

be minor differences for example in the size and shape of a button. The red color in figures indicates that the 1177 direkt had a shade of red, but the colors are not exactly the original colors. The conversational agent's dialogue box, the chat bubble, is grey in our figures but the original color in the 1177 direkt is a shade of light pink. In this paper, a black border has been added around the figures to clearly define the edges. The alterations in the figures, are made to avoid copyright violations but still show the main aspect of detected design issues. The text in the figures is translated from Swedish to preserve the original meaning or essence. The figures are at times presented with fictitious text, reflecting content in the 1177 direkt.

4.1 The design of the eHealth service created expectations that were not met during interaction

This theme presents the participants' unmet expectations when using the eHealth service. The participants' expectations on the chatbot were influenced from both design features and from the participants prior knowledge of the web-portal 1177.se, chatbots, and AI.

The workshop participants from the Begripsam group perceived 1177 direkt as part of 1177.se due to the similarity in name and design and therefore expected 1177 direkt to have access to their existing patient healthcare records and other data in 1177.se.

'I just assumed it would be compatible with the regular system.' (P1).

Accordingly, the participants expressed that the question about diagnosis (Figure 2) should have been automatically retrieved from the patient's healthcare record in 1177.se.

The workshop participants from the Begripsam group expected the 1177 direkt to be easy to use since it was presented as a chatbot, with both an icon, and a word for chatbot (Figure 3).

The layout also indicated that a chat conversation was taking place, with the chat dialogue placed at the bottom of the screen

(Figure 3). However, the interaction did not function as expected. Instead of building a conversation from bottom and up with a possibility to scroll to see the previous dialogue, the user was redirected to a new page after each question.

'I take it that since it's all white up there, I figure it's going to be like a chat, you know, moving upwards.' (P4).

'It's unreasonable because when you think of a chat and [the term] immediately, you think it should be quick and smooth. I think it's way too much.' (P3).

Workshop participants from the Begripsam group criticized that the 1177 direkt was presented with misleading design and terminology.

'I also thought about how this chatbot feels like a questionnaire. I mean, why don't they just call it a questionnaire from the start? Like a form with yes/no questions? Instead of a chatbot. It's also easier for the older generation.' (P3).

This was in line with the paired interview participants.

'I think the expectations are lower if I tell a patient right from the start: - Here, you don't have anyone to chat with, here's a form. If you fill it out, you'll get to the right person.' (P18).

Workshop participants from the Begripsam group expected the chatbot to rely on AI and use prior responses to inform subsequent queries, thereby eliminating redundancy, but this was not the case. The term 'epidural' was explained as a type of anesthesia that is often administered during childbirth.

'We've already said before that we haven't been pregnant or given birth in the past few weeks, so we couldn't have had this epidural either. It's unnecessary to bring it up now.' (P3).

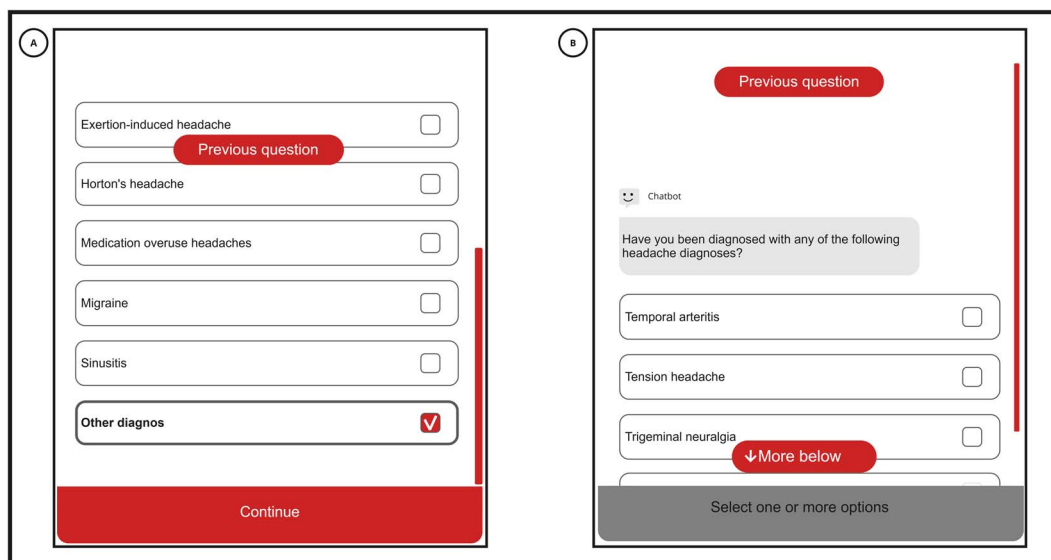


Figure 2. A) The figure illustrates the end of the first page for the question about previous headache diagnoses. Options are listed alphabetically, with 'Other diagnoses' as the last option. Selecting an option, such as 'Other diagnoses', changes the interactive element at the bottom from grey to red, shifting the text to 'Continue' as shown here. The floating action button 'Previous question' is a link to the previous page and partially obscures text when scrolling. B) The figure illustrates the beginning of the second and last page with a list of options for the question on previous headache diagnoses. The interactive element at the bottom is disabled, presented in grey color with instructions to select one or more options.

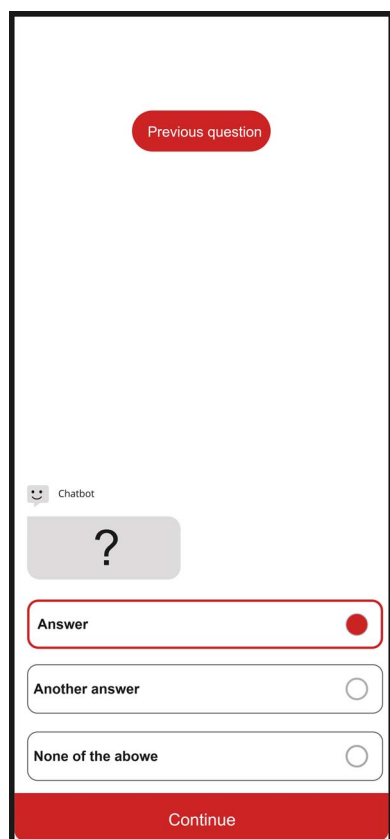


Figure 3. The chat is placed at the bottom of the screen. The word Chatbot and an icon for chatbot are presented above the question. The question is represented by a question mark.

A general feeling expressed by participants was that they expected the 1177 direkt to be advanced, incorporating the latest technologies.

'We have come so far in technological development in society in general, that we're spoiled as citizens, as people, with apps that can, like, start our cars remotely and all sorts of things.' (P17).

It was not possible to seek help for another person, except for children in a certain age under one's legal guardianship. However, workshop participants from the Begripsam group expressed expectations for this functionality, since it is possible to call 1177 for consultation on behalf of someone else.

4.2 The eHealth service uses a difficult language and a lot of text

This theme relates to experiences regarding the proficiency level of the text that is not adapted for users with lower levels of literacy, cognitive load, value-loaded expressions, readability, and understandability.

The chatbot builds on content from a triage manual used by nurses conducting triage of patients calling the phone number 1177. According to the participants in the paired interview, the medical terms used in the manual informs the nurses, but the nurses adapt their language to the patient.

'But if you use the same terms and concepts with a patient, they won't understand it.' (P18).

Workshop participants from the Begripsam group found many words hard to read, causing problems with the whole triage and symptom check process.

'You notice that it is difficult for me to read that word, the first word, then I am kind of immediately lost.' (P4, trying to read out loud the Swedish word for whiplash-injury.)

One question targeted if the user had undergone a 'procedure', which was perceived as an uncommon word that was hard to understand by the workshop participants. The explanation of

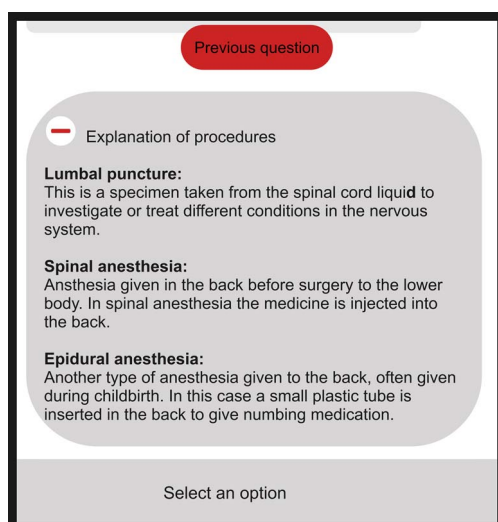


Figure 4. The expanded information box contains explanations of lumbar puncture, spinal anesthesia, and epidural anesthesia. The text contains uncommon and long words, long sentences, and medical terms.

'procedure' consisted of explanations of lumbar puncture, spinal anesthesia, and epidural anesthesia that included uncommon words, medical terms, long words, and long sentences (Figure 4). When trying to understand the word 'procedure', one workshop participant from the Begripsam group forgot the question and burst into excessive laughter. This can be interpreted as too much cognitive load.

One question addressed if the headache started after injuring or twisting the neck. The explanation provided an example of pain after 'chiropractic or naprapathy manipulation'. The word manipulation had a negative connotation for P2, hence also the treatment. The participant had previously been treated with the mentioned 'manipulation'.

'...but using the word manipulation, off the top of my head it's some kind of bad treatment.' (P2).

Workshop participants expressed that the readability of the text affected the understanding of the information and questions. The use of dropdown lists, small fonts and how information was presented made it hard to read text in the service e.g., a long list of examples in parenthesis (Figure 5).

4.3 The eHealth service triggers negative emotions and reactions

The workshop and paired interview participants expressed emotions triggered by the eHealth service that may affect people in a negative way.

A common emotion expressed by workshop participants was a feeling of tiredness or exhaustion.

'I also suffer from brain fatigue, which is a diagnosis too, and it makes you incredibly tired in the head. I mean, if I had the energy to get this far [to step 16] when I was seeking help for headache, then I would have hung up and called my primary healthcare center on the phone instead.' (P2).

The many steps in the 1177 direkt caused frustration and felt overwhelming according to the workshop participants. One participant pointed out that a healthcare professional would not

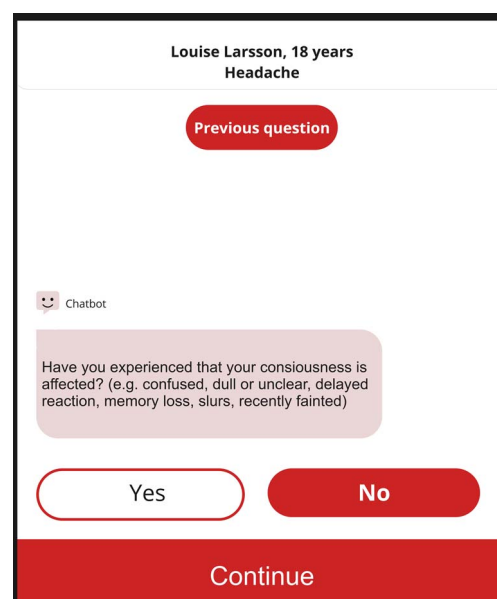


Figure 5. The question is presented in a pink chat bubble, followed by a long list of examples in parenthesis.

ask so many questions and expressed in an angry voice that the instructions were unnecessarily detailed.

'The way it's phrased there, like, 'Have you measured your body temperature with a thermometer?' First of all, I'd be like, what? Annoyed by the question. What else would I have measured it with? Like, should I have used a hammer.' (P2).

Workshop participants from the Begripsam group expressed hesitations about using the system and reflected on that they would have probably interrupted the triage and symptom check process if they had tried 1177 direkt in a real situation-seeking healthcare.

One workshop participant expressed a concern about the risk of providing incorrect answers and not being able to correct them.

'My reaction is: I think I might see a mistake, but I don't bother fixing it because I just don't have the energy.' (P12).

The participants in the paired interview reported that healthcare professionals experienced stress and an increased workload. This was due to uncertainties about whether they had correctly understood the patient's answers presented for healthcare professionals as a generated summary text based on the patient's answers, and concerns about whether the patient had accurately understood the questions.

'And they [the healthcare professionals] are feeling a little worried. And then they maybe ask [the patient] some additional questions just to be super sure that they really understand each other now.' (P17).

Additionally, this raised concerns about misinterpreting the generated summary text potentially creating patient safety risks and fears that this could lead to personal consequences.

'Then we have the next part, the one about it [the generated text] creating uncertainty, which sometimes leads to questions like: "Can I lose my [healthcare practitioner] license?" from our nurses. This concern is based on worry.' (P17).

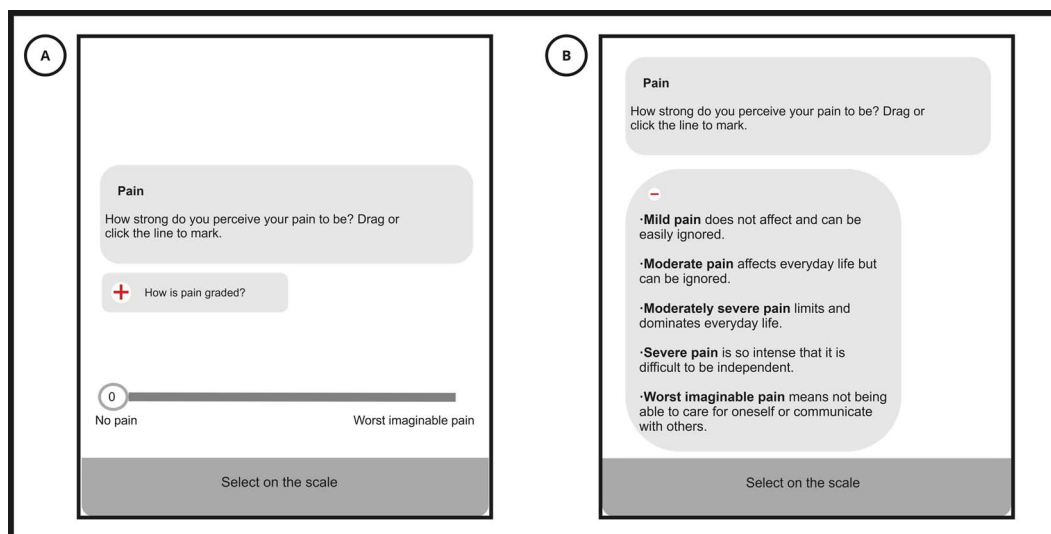


Figure 6. A) Pain scale with a movable marker starts at 0 ('no pain') and ends at 10 ('worst imaginable pain'). The number changes as the marker moves. B) The expanded information box with explanations for pain levels: mild, moderate, moderately severe, severe, and worst imaginable.

4.4 The layout and graphic design of the eHealth service is difficult and sometimes misleading

Several challenges were identified in the layout and graphic design by the workshop participants, as presented in this theme.

The lack of information about progress was experienced negatively and could present a barrier for the workshop participants to complete the triage and symptom check process.

'I also feel like it's more of a questionnaire, and it would be right to have it like we said before, where you have one, two, three, four, five, and you know you're done with one, then two, then three, then four. Here, you don't know how long you're supposed to keep going. I find that a bit annoying.' (P4).

The 1177 direkt is heavily based on written text. The workshop participants from the Begripsam group expressed the text-based design as hard to use and looked for an alternative modality such as text-to-speech, but there was none.

'Because I think there are lengthy words and such. And the question is, I just thought of this now, can you listen to it [the text]?' (P4).

The 1177 direkt required that the user have a high capacity of remembering. The question addressing the previous headache diagnosis provided a list of options with the diagnosis lacking a response option indicating uncertainty, for example 'I don't know'. One workshop participant stated that even if they have been previously diagnosed, this does not automatically mean remembering the correct medical term for the diagnosis.

Workshop participants from the Begripsam group expressed that self-rating scales in healthcare are generally hard to use and that the presented self-rating scale for pain (Figure 6) was hard to understand. It consisted of a 11-step scale where users select a value by moving a numbered circle to the desired position (Figure 6A). An expandable information box marked with a cross (Figure 6B) provided an explanation about grading the pain as followed:

- Mild pain does not affect and can be easily ignored.

- Moderate pain affects everyday life but can be ignored.
- Moderately severe pain limits and dominates everyday life.
- Severe pain is so intense that it is difficult to be independent.
- Worst imaginable pain means not being able to care for oneself or communicate with others.

One participant expressed that the explanations were very good. However, the mismatch between the horizontal line's eleven steps and the explanation's five steps along with lack of guidance caused difficulty understanding how to select a value.

'These options must, sort of, be the same. They must be connected, the line with the suggestions. Otherwise, are we supposed to count that the first [explanation] is the number one [on the line] and the second is the number two, right? There must be some logic to it.' (P5).

The chat provided different kinds of interactive elements for options: radio buttons (Figure 7A); pre-enabled button acting as a link (Figure 7B); buttons with different options presented side by side (Figure 7C); checkboxes (Figure 7D); and clickable fields (Figure 7E).

The lack of consistency in the applied interactive objects created confusion and uncertainty on how to act.

'I just wanted to point out that I don't feel good about them changing the functions. Before, the yes button was buttons, and now it's all white with a red dot on the right side. It can be confusing that there are different functions for what you're supposed to press. You might be unsure if you should press the round ring, press there, and now when it only shows red on the side, people might get unsure because it didn't turn completely red.' (P3).

The red color of the continue button sent mixed signals regarding the appropriate action to take. The label on the button said Continue, but the red color was regarded as a stop or warning signal (Figure 7E).

'I think red is more like something alarming or a stop signal. Green is more inviting.' (P2).

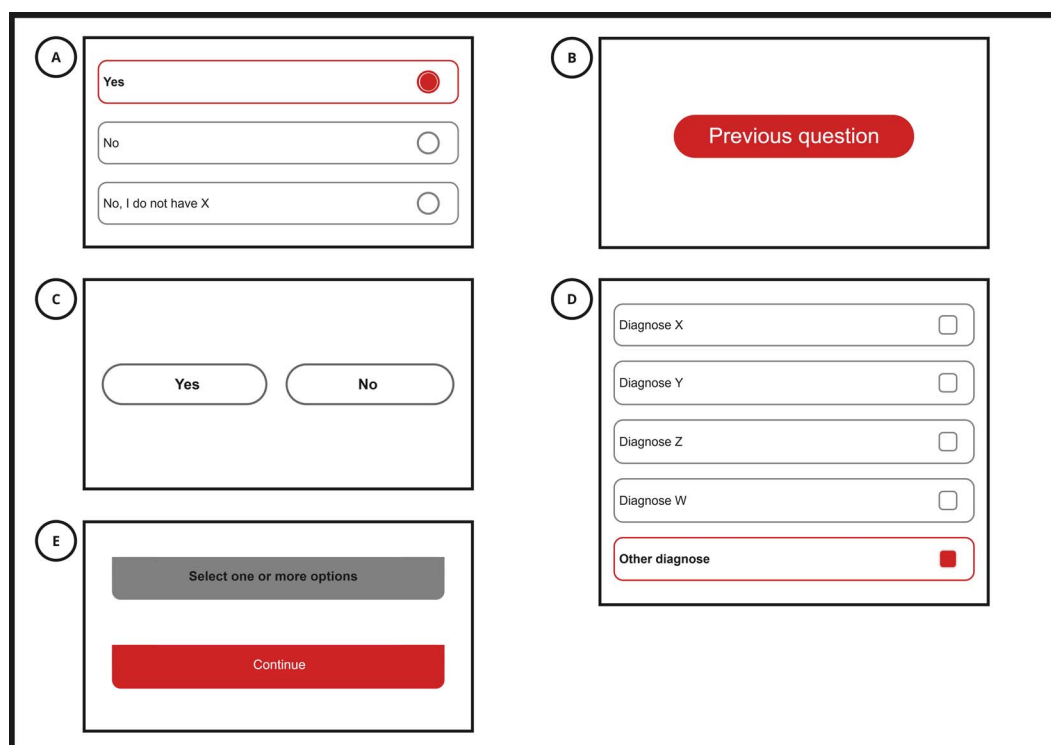


Figure 7. A collection of interactive elements in 1177 direkt, A) radiobuttons with a list of options, yes is selected, B) pre-enabled button that act as a link to the previous page, C) clickable buttons side by side with different options in each button, D) a list with options provided in checkboxes. 'Other diagnoses' is selected, E) clickable fields, placed at the bottom of the screen. When inactive, they were grey (e.g., 'Select one or more option'). Upon selection, they turned red and became active links with new instructions (e.g., 'Continue').



Figure 8. Magnifying glass with a heart in the middle, which was used as an icon for a search box to enter a symptom considered confusing by workshop participants.

Prior to chat entry, users selected their medical condition by typing in a search box. The search box's heart-inscribed magnifying glass (Figure 8) was deemed confusing.

'Why is there a heart in the middle? I get that it's a magnifying glass, but I don't understand what the heart is doing there.' (P6).

The background color made it difficult for some to read the information (Figure 5).

'I just think that this pink background makes it hard to read the text. I find it difficult to read.' (P7).

4.5 The eHealth service lacks in trustworthiness

The participants in both the workshops and the paired interviews reported a loss of trust in the eHealth service.

The paired interview participants raised trust issues relating to the system's inability to account for a patient's previous illnesses or multiple symptoms. By focusing only on the first symptom entered, the triage capabilities are limited and lack a holistic view of the patient.

'And this makes patients feel like, 'I have a sore throat, I'm coughing, I have pain in the ear'. There can be several things the patient wants to communicate, but they can't. They have to choose one symptom, and that's what counts.' (P17).

The design of the user interface limits patient communication, as it does not provide options for open-ended responses or for expressing uncertainty. The workshop participants considered it important to be able to express uncertainty considering the wide range of information a patient may not be aware of or remember, including, but not limited to, the previously mentioned medical term for their diagnose in theme numbered 4. Paired interview participants highlighted that the limitation in the communication with pre-defined questions and response options created a risk of compromising a comprehensive reporting on health issues. According to the paired interview participants this concern had been reported to them by both patients and healthcare professionals. The questions and options did not always cover the patient's health issues or concerns.

'There were hundreds of thousands of questions, but I couldn't express what I wanted.' (P18).

Trust was negatively affected by a design that was experienced as illogical, hard to use or inconsistent. When reaching step 14

(out of 25), the 1177 direkt asked 'Have you experienced that your consciousness is affected?' together with a list of examples including confused, lethargic or unclear, delayed reaction, disoriented, memory loss, slurring and recently fainted (Figure 5). The participants got upset about this and the nearest following questions that were identified to determine if the patient may have suffered from stroke or other serious illness. Several of the participants reacted strongly, pointing out that the questions were positioned far too late in the process.

'Several of these questions feel like they should have been asked already, like they've already been sorted out. Like, is your awareness affected? If I'm experiencing memory loss, feeling disoriented, or fainting, that should be something I would have chosen to mention right away.' (P2).

Workshop participants from the Begripsam group questioned if a patient suffering from disorientation would accurately respond to questions about their condition, highlighting the absence of a direct line to healthcare professionals.

'Maybe there should be a button here for if I'm about to faint or feeling drowsy, or if something's happening, so you can press a button to call 112² or something.' (P4).

The 1177 direkt ended the triage and symptom check procedure of the fictive health condition with a recommendation to book an appointment with a healthcare professional in the eHealth service. The booking should be completed within seven days according to the instructions. This was interpreted by the participants as confirmation that it was not an urgent matter.

'After all these questions, you might be completely exhausted, and they [the chatbot] say it's not urgent [...] It [the chatbot] said that by the sixth at the latest, you can see a healthcare professional digitally.' (P3).

In addition, the workshop participants expressed concern that users might struggle to book the recommended appointment with a healthcare professional if they postponed the booking, especially without reminders and booking support.

4.6 Suggestions of improvements and reflections

The fourth workshop was dedicated to idea generation aiming to improve cognitive accessibility. A detailed list of the suggested solutions from workshop 4 related to barriers, themes and chapters in the cognitive accessibility standard are presented in Appendix 2.

At the beginning and end of the workshop, questions were posed to encourage reflection on the insights gained during the sessions and how the insights would be utilized. Representatives from both product owner perspective affirmed the value of participating. The representative from Platform24 explained that they had made a report from the first three workshops and already informed others in the company. The representative from Inera explained they wished to implement similar methods into their processes.

5 Discussion

In this study we have explored the user's experiences of the conversational agent 1177 direkt by focusing on the design of the user interface and on cognitive accessibility.

Our study may be the first to engage both participants with cognitive impairment and participants from the product owner in the workshops to both evaluate and to generate ideas for improved cognitive accessibility. We consider our method to be a novel, building on PAR and PD aiming to affect the design decisions when the ownership of the system is outside the research group.

We identified several design- and language-related accessibility barriers that affected how participants understood the questions and how they responded. We believe these accessibility barriers may cause a risk of incorrect responses and therefore the triage may not be accurate. These findings are in line with the results from the user testing study of two symptom checkers in Finland, where participants with trouble of understanding the questions in the symptom checkers added symptoms that affected the AI-based assessment of the urgency (Savolainen and Kujala, 2024). Previous research has pointed out that digital and online symptom checkers have lower accuracy compared to healthcare professionals (Chambers et al., 2019) and other comparisons (Wallace et al., 2022). Considering the risk of incorrect responses, we propose future research to investigate relations with accuracy and design in all kinds of digital or online symptom checkers and triage tools.

Madanian et al. (2023) argue that there is a need for patients, clinicians, and technicians to work together when designing eHealth to overcome barriers for using eHealth. Our paper presents a case study that contributes important knowledge to HCI by showing how stakeholders can employ cooperative strategies in the evaluation and co-design of eHealth services to obtain a rich data material and a reliable data analysis. However, the shown cooperative strategy should be implemented early in the design development process to avoid fundamental design issues that may be too costly to edit afterwards, e.g., presenting the 1177 direkt as a chatbot when it is experienced as a questionnaire. The early approach is supported by the standard Ergonomics of human-system Interaction ISO 9241-210 that state 'early evaluation are likely to be the most cost-effective' (International Organization for Standardization, 2019).

It is important that evaluations and co-design processes for accessibility are conducted with end-users who have lived experiences of disability due to cognitive impairments in collaboration with accessibility experts that have sufficient UX knowledge to avoid proposing a design that goes against best practice. This approach and the positive effects for organizations has been described by Johansson et al. (2023). In our study a participant suggested underlining text for focus during one of the evaluation workshops, but underlined text should be used exclusively for links on webpages and in apps. The accessibility expert explained this during the workshop, an example of the mutual learning process in PAR.

Both the summary of chatbot accessibility by Stanley et al. (2022) and the ISO 21801-1:2020 guidelines on cognitive accessibility point to the importance of context (International Organization for Standardization, 2020). The ISO guidelines on cognitive accessibility emphasize that design that draw on users' prior knowledge and experiences may be easier to use (International Organization for Standardization, 2020). However, the standard lack specific guidance on which kind of knowledge and experiences could affect cognitive accessibility. The knowledge may be

² 112 is the Swedish emergency telephone number.

context dependent, as shown in our study. The context dependence in our study related to participants' knowledge on how the Swedish healthcare functions and is organized. We also found that participants' knowledge and experiences about the national eHealth web-portal 1177.se, chatbots and AI were important and affected participants' experience of the 1177 direkt. The national eHealth web-portal 1177.se is a digital ecosystem, and according to Inera it is a strong brand (Inera, 2023). However, few of the services in the digital ecosystem exchange information. We found that participants were unaware of the lack of integration and lack of information exchange within the digital ecosystem in 1177.se, leading to unmet expectations and negative feelings. We criticize the lack of information for users that the components in the digital eco-system 1177.se operate as standalone services without effective information exchange and integration. Our result shows that designing the conversational agent with the graphical design from 1177.se is very problematic. If integration and information exchange is not possible, the result from the idea generation workshop suggest the conversational agent should be presented with a distinctive name and graphical design to show it is a standalone eHealth service. We recommend eHealth service developers to clearly describe the functionality of the conversational agent to establish reasonable expectations for the user. This is in line with the summary of chatbot accessibility by Stanley et al. (2022). In addition to the accessibility guidance in the summary, we suggest that patient portals with digital ecosystems should be designed to clearly indicate whether a specific system within the digital ecosystem facilitates integration and information exchange. We suggest avoiding similarity in graphical appearance and names for systems that lack integration and information exchange.

The theme 'The design of the eHealth service created expectations that were not met during interaction' shows the importance in avoiding using design metaphors that promise more than the eHealth service delivers. This is shown in the result around presenting the 1177 direkt as a chatbot when actually it is functioning as a questionnaire. Also, the unmet expectations on advanced AI in this theme may mirror the recent years societal focus on AI, as well as the increasing presence of AI tools in everyday life. Adamopoulou and Moussiades (2020) highlight the growing research interest in conversational agents, presenting a wide range of concepts and technologies in this field. The variety of technology in conversational agents and increasing presence of conversational agents in daily life (McTear et al., 2016) may contribute to users having different experiences of using conversational agents. Another explanation may be media framing AI as a person and replacing or outperforming the human medical expert (Bunz and Braghieri, 2022). Our result shows that participants expected healthcare to implement advanced technology in a conversational agent, instead of technology like already existing questionnaire services in 1177.se that participants had prior experiences of.

The theme 'The eHealth service triggers negative emotions and reactions' shows a spectrum of negative emotions and reactions triggered by the 1177 direkt, relating to perceived stress when testing the 1177 direkt. Given the harmful effects of stress (Yaribeygi et al., 2017), and that stress can affect the user's performance, it is crucial to design eHealth services that mitigate such negative responses. We highlight the importance of designing conversational agents together with end users with lived experience of disability due to cognitive impairment to meet user needs and expectations and to avoid triggering negative emotions and reactions. The participatory approach is important to investigate

what prior knowledge and experiences must be considered when developing a conversational agent.

The theme 'The eHealth service lacks in trustworthiness' shows a potential risk for patient safety at the end of the triage and symptom check. In our evaluation the 1177 direkt recommended to book an appointment with a healthcare professional. The appointment could be booked in a timeframe of seven days. One participant thought the span of seven days indicated a non-urgent health issue. However, the timeframe was not related to the priority or urgency of the specific health issue. The time frame is set by the healthcare provider for administrative reasons, regulating the time frame a patient can access the available appointments presented in a calendar. Even if a case is not triaged to acute medical attention, waiting seven days could be potentially harmful for the patient, prolong the period for recovery or increase the cost for healthcare interventions. Hence, also non-acute issues should have different levels of urgency and thus should not be given the same timeframe to respect the principle of need and solidarity.

EU regulations (Directive (EU) 2016/2102) require public eHealth services to be understandable (European Telecommunications Standards Institute: ETSI, 2018; World Wide Web Consortium, 2018). The eHealth service we investigated targets the whole population needing primary healthcare in the regions that implemented 1177 direkt. However, complex language and extensive text can be disabling for many users. According to the National Agency for Special Needs Education and Schools 20%–25% of the Swedish population have reading and writing difficulties due to various reasons, such as other native languages, neuropsychiatric diagnoses, cognitive impairments, dyslexia, and sensory impairments (The National Agency for Special Needs Education and Schools, n.d.). The Survey of Adult Skills (PIAAC) (OECD, 2012) reports that while Swedish adults (16–65 years) have above-average literacy skills, 42% of the population have skills up to level 2. We found the text in 1177 direkt to be at least level 3 based on the explanations of levels in the PIAAC. 1177 direkt requires users to understand and navigate complex digital texts, identify, interpret, and evaluate one or more pieces of information (e.g., explanations of uncommon words). Further on, the 1177 included multi-step operations (i.e., list of options divided on two pages), and required users being able to disregard irrelevant or inappropriate content to answer accurately (i.e., long lists of multiple-choice answers) and included competing information (i.e., two questions in one). This complexity means 42% of the population may struggle with 1177 direkt. Some content even reaches level 4, involving uncommon medical terms and requiring users to identify specific, non-central ideas. Only 16.1% of Swedes have literacy skills at levels 4 or 5. We recommend simplifying the language in eHealth services to meet the needs of the Swedish population. Considering the literacy distribution in all countries reported in the PIAAC, healthcare conversational agents worldwide must use carefully formulated easy to read and easy to understand content to be accessible and beneficial to all users.

Pictures and symbols can support readability (Friedman and Bryn, 2007; Williams and Hennig, 2015; Yaneva et al., 2015) and the user's understanding of the system. However, adding pictures or symbols that do not make sense can make the user confused, as was the case with the heart icon in the magnifying glass (Figure 8) indicating the search bar in 1177 direkt.

The CoDeAc research program that conducted this study has a background in PAR aiming for change. Therefore, we need to reflect upon on if the suggested improvements could have real impact on the design of the eHealth service. In our opinion, several

of the suggested solutions should be easy to implement, such as editing text to be understandable for a broader population. The language used should be neutral, avoiding words associated with negative meaning. Other suggestions are that existing assistive technology should be implemented into the system, for example text to speech technology and speech to text technology. This is in line with previous research (Huq et al., 2024). Other improvements could be more complicated, but nevertheless important. Integration with information exchange between 1177 direkt and the patient's healthcare record could provide important patient information without having to rely on the patient's memory. Also, allowing representatives in a personal eHealth service could improve the service. This is not yet regulated by Swedish law³, indicating that there might be need for regulatory changes to create an inclusive healthcare ecosystem where relatives can play an important role. This is in line with demands from the WHO (World Health Organization, 2011).

It is important that triaging eHealth services are designed so a very broad range of users can use them, respecting the CRPD (United Nations, 2006) and in Sweden also the principles in the ethical platform (Hälso- och sjukvårdslag, SFS, 2017:30) (SOU 1996/97). Hence, design with cognitive accessibility and high levels of usability in mind should be imperative. The procurement agreement regarding the platform holding the tested system is reported to be worth one billion Swedish kronor (Caesar, 2022) which is approximately 88 million Euros. Research indicates that people with disabilities consume more healthcare (Rodríguez et al., 2024) and the WHO reports that people with disability have poorer health compared to people without disability (World Health Organization, 2022c). Therefore, it is unacceptable that people with impairment find eHealth services being inaccessible. Reports from Pettersson et al. (2023) show that especially people with difficulties with reading, writing, calculating, and communicating alongside with people with intellectual difficulties report eHealth as being inaccessible for them. For eHealth services to be cost effective, they need to be used by the widest possible range of the population and especially by the part of the population who need to utilize healthcare the most. We argue that designing cognitive accessible eHealth services beyond the present legal accessibility requirements in the EU is necessary to reach the EU's vision of a Union of equality. As previously mentioned, following EN 301549 or WCAG 2.1 on AA level is not enough to design cognitively accessible eHealth services. People seeking healthcare may have several conditions or be in situations that affect cognition. For people older than 50 years living in the community, the global prevalence of cognitive impairment is reported to be between 5.1% and 41% with a median of 19% (Pais et al., 2020). People may also suffer from temporary or chronic conditions that have a negative impact on cognition, for example, hyperthermia (Walter and Carraretto, 2016) or pain (Moriarty et al., 2011). Supported by our study we suggest that cognitive accessibility can be enhanced by actively involving people with cognitive impairment in the design process together with the application of the general guidelines on cognitive accessibility ISO 21801-1:2020 (2020).

We argue it is important that eHealth services maintain the CRPD (United Nations, 2006) and the ethical platform regulated by Swedish law (Hälso- och sjukvårdslag, 2017). As discussed above,

there is a risk the CRPD (United Nations, 2006) and the ethical platform principles for (i) human value, (ii) need and solidarity and (iii) cost-effectiveness are not followed due to the design of the tested 1177 direkt.

5.1 Strengths and Limitations

A strength of the study is the constellation of the evaluation group, with participants with lived experiences of disability collaborating with the multidisciplinary researcher group and product owner representatives. As previously shown by Johansson et al. (2023), participation from a group like the Begripsam group bring valuable knowledge from their lived experience as experts from the extreme user perspective. The participants from Begripsam group were experienced in the method of evaluation and co-design from a cognitive accessibility perspective and have lived experience of healthcare. The researchers bring their expertise in accessibility, design and primary healthcare and the product owners bring their knowledge about the system and considerations behind the systems final design. The collaborative approach in the spirit of PAR (Macdonald, 2012) and the evaluation process with the barrier walkthrough could be used to identify a wide range of cognitive accessibility issues. Together we experienced first-hand narratives of potential service users. These narratives authentically convey a holistic chain-of-events (also tacit and layered) hampering cognitive accessibility that is in line with user involvement as proposed by (Beresford, 2013; Greenwood and Levin, 2007b). Further, dialogues produce mutual discovery and insights of barriers, which may enlighten participants with potential practical solutions for all. In other words, collaborative methodologies are needed to take on more complex challenges than the positivistic paradigm (Greenwood and Levin, 2007a). Everyday experience is often the missing ingredient in development and is at focus in PAR (Macdonald, 2012), with the possibility to find neglected needs, which need after-the-fact redesign.

The participants with lived experience of cognitive impairment were all native Swedish speakers. Therefore, we did not investigate any barriers related to foreign language or cultural barriers. The participants are internet users and have experience as patients in healthcare, so there may be more barriers that we did not detect, that could be experienced by people who seldom or never use the internet or rarely contact healthcare in Sweden. The PCBW was conducted in a group format, which may have caused that not all considerations were lifted by the participants despite our efforts to create a creative and safe workshop environment. Nevertheless, we believe that discussions in a group will provide for a richer and deeper understanding of potential problems related to the service than is the case when using individual test settings.

There are limitations in our study. There are practical requirements to use the system such as having access to a mobile phone number and electronic identification; however, these requirements are outside the scope of this study. The PCBW was conducted in a training module that mirrors the implemented 1177 direkt but may have some differences that we are not aware of. Our focus was on cognitive accessibility. People with other impairments might have identified additional barriers. The collective experiences of the participants from the Begripsam group ranged broadly over cognitive impairments but did not cover all cognitive impairments, e.g., intellectual disability. The remote setting of the workshops may pose limitations; however, the participants from the Begripsam group had previous experience of both the method and the remote setting.

This study is situated in the public healthcare context in Sweden, but we believe that several of the results are transferrable to

³ The proposition regarding representative in eHealth services has not been decided. In personal email query from first author to the Government Office in May 2023 regarding 'SOU 2021:39 Ombuds tillgång till vård-och omsorgsuppgifter och förenklad behörighetskontroll inom vården' the Government Office answered that the Government had not yet decided when to process the proposition. Updated questions were filed on May 15, 2024 by the first author.

other contexts. For eHealth services to be experienced as accessible for the greater part of a population, it is important to design with universal design principles (Story et al., 1998). This includes acknowledging language as a vital part of the design catered to different literacy levels. Our result shows the user's background knowledge and experiences should be taken account for in all eHealth service development and the participatory approach provides well for the purpose. Presenting a service as a chatbot sets user expectations. If user expectations are not met, it may trigger negative emotions and reactions. This finding is likely applicable in other contexts as well.

5.2 Future research

More research is needed to investigate how patient portals with digital ecosystems should be designed to be experienced as cognitively accessible and meet the user's needs. Further studies from our research group will focus on this area.

Since conversational agents is a relatively new area and with increasing involvement of AI in the services it is important to continue to do research on how this technology include or exclude people with impairments. It is also important to study if this technology introduces or enhances biases and discrimination based on notions of ableism or 'othering'.

5.3 Conclusion

In our study, we explored the design of a conversational agent implemented in nine regions in the Swedish national web-portal for health information and services available for residents in the specific regions. The result shows that (1) the design of the conversational agent affected the user's expectations, (2) the language is a vital part of the design that affected experiences of cognitive accessibility, (3) the eHealth service triggered negative emotions, (4) layout and graphical design affect user experience, and (5) negative experiences created trust issues. The workshop dedicated for idea generation shows how participants with lived experience of cognitive impairment provide valuable suggestions to improve cognitive accessibility such as providing options for expressing uncertainty.

It is important that eHealth services in trusted systems serve as a model for inclusion. So far although, conversational agents have introduced yet another arena where the lack of accessibility divides people into those who can or cannot use the services. If designed and implemented correctly, this technology has the potential to improve healthcare and provide for a more inclusive society.

The EU regulation (Directive (EU) 2016/2102) and the WCAG 2.1 (World Wide Web Consortium, 2018) focus on visual impairments and assistive technology, therefore a vital part of accessibility is missed. To create accessible digital services that works for all people, cognitive accessibility must also be considered. We argue that it is crucial to include people with lived experience of cognitive impairment in the evaluation and development of eHealth services to ensure these services benefit everyone.

Supplementary material

Supplementary material is available at *Interacting with Computers* online.

Statements and declarations

Conflict of interest: MJ, LP, and CG work in the public healthcare sector, Region Västra Götaland and Region Dalarna, where the investigated eHealth service has been implemented. The

employers' organizations had no role in the undertaking of this study.

Data availability

The data underlying this article cannot be shared publicly due to privacy concerns for the individuals who participated in the study. The data will be shared on reasonable request to the corresponding author.

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